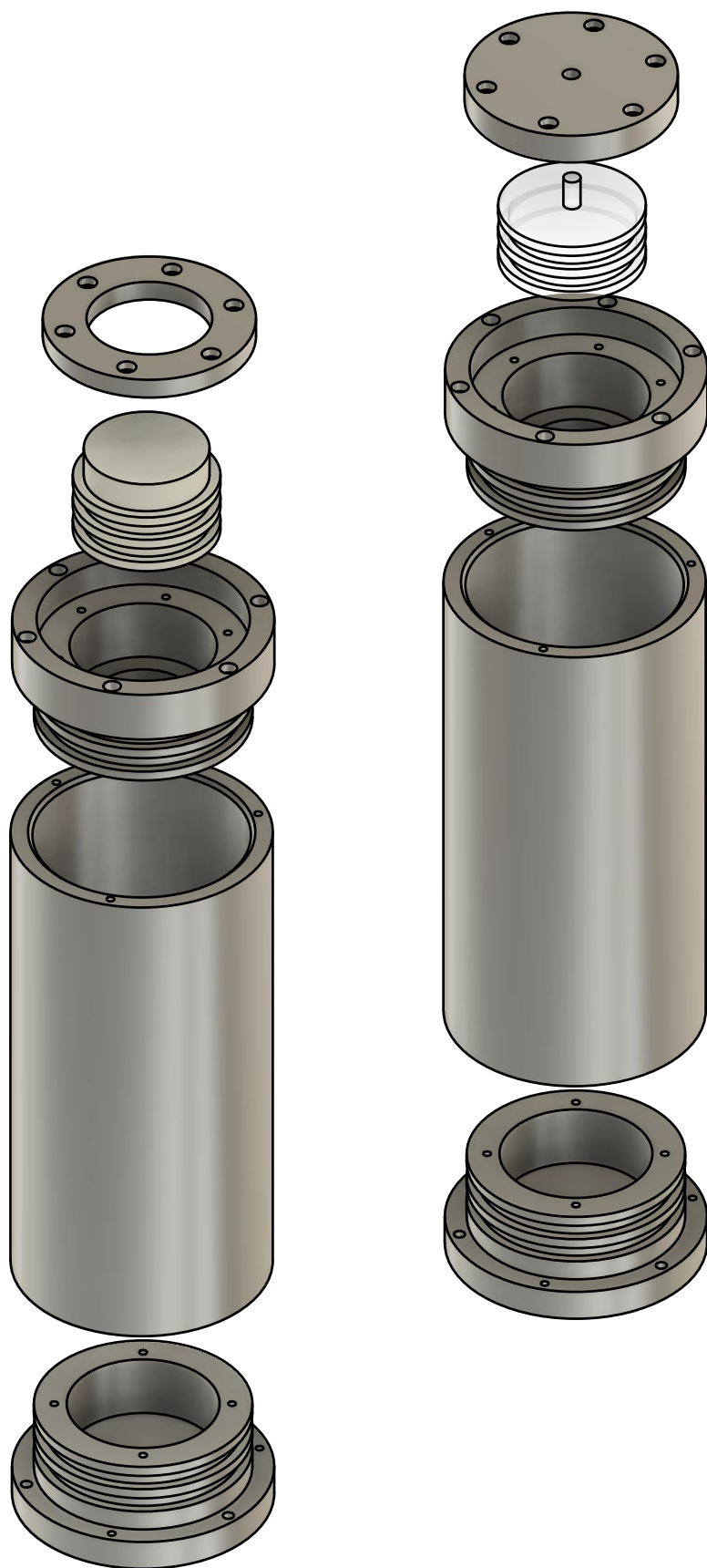
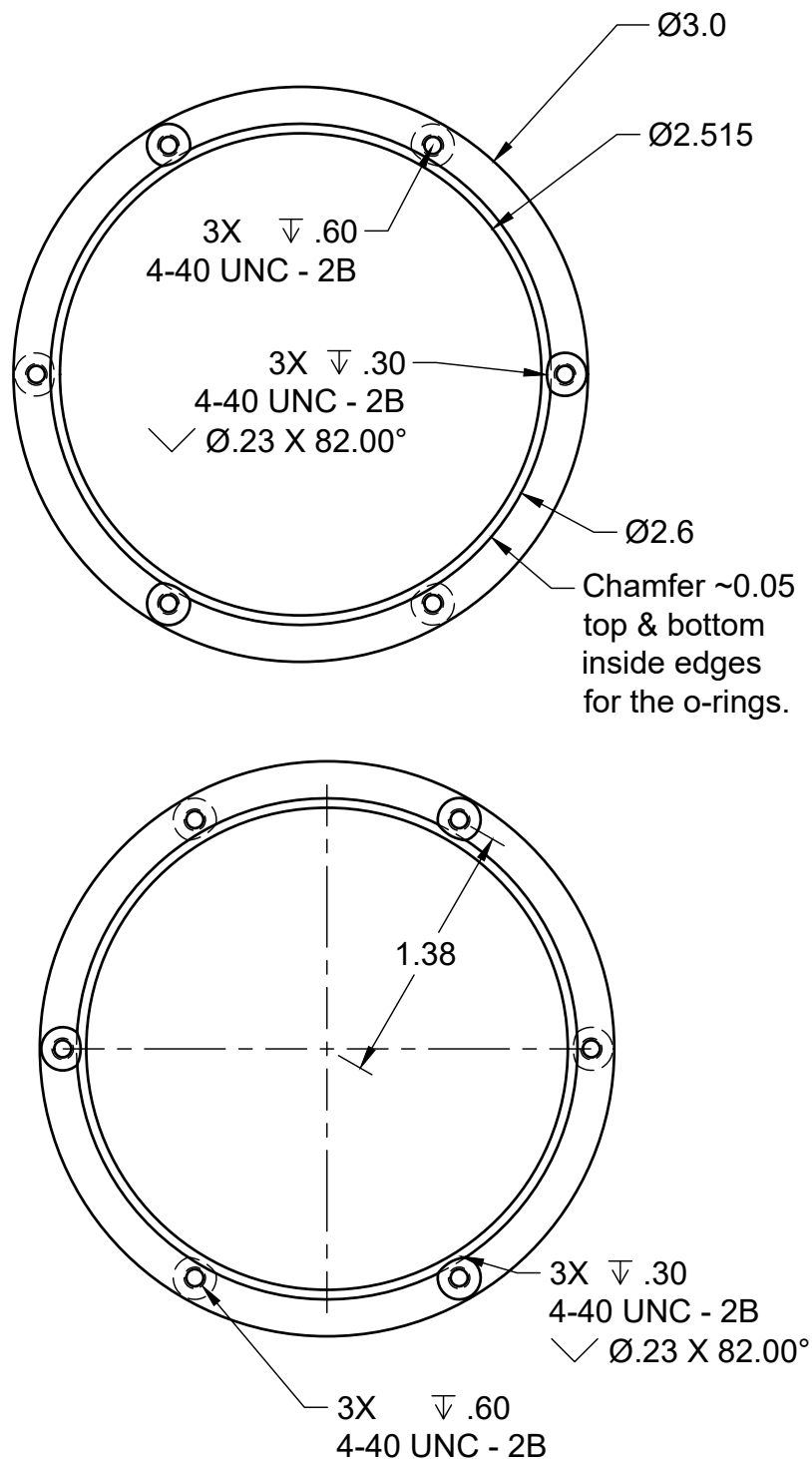
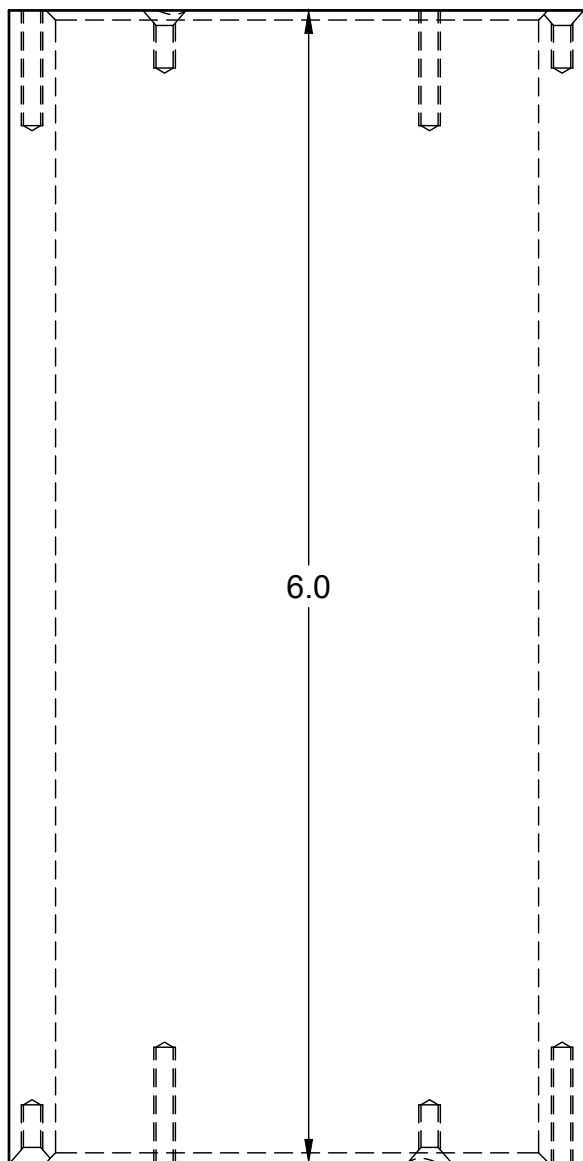




Rev 4: "special" Flat head 0.225" Ø
MMC #: 94063A063
& have an undercut profile.



AL Tube: 3" Ø, 1/4" wall, 6" Length.

3x Top holes +0.6" deep
3x Bottom holes +0.6" deep
1/4" Flat head (6x) ~0.3" deep

[top & bottom taps do
not need end to end alignment.]

PROJECT

BHR

TITLE

AL Tube: 3" Ø

Rev 4: 2nd set of threaded holes for Flat Head "stop" screws

APPROVED

CHECKED

SIZE

A

CODE

DWG NO

REV

4

DRAWN

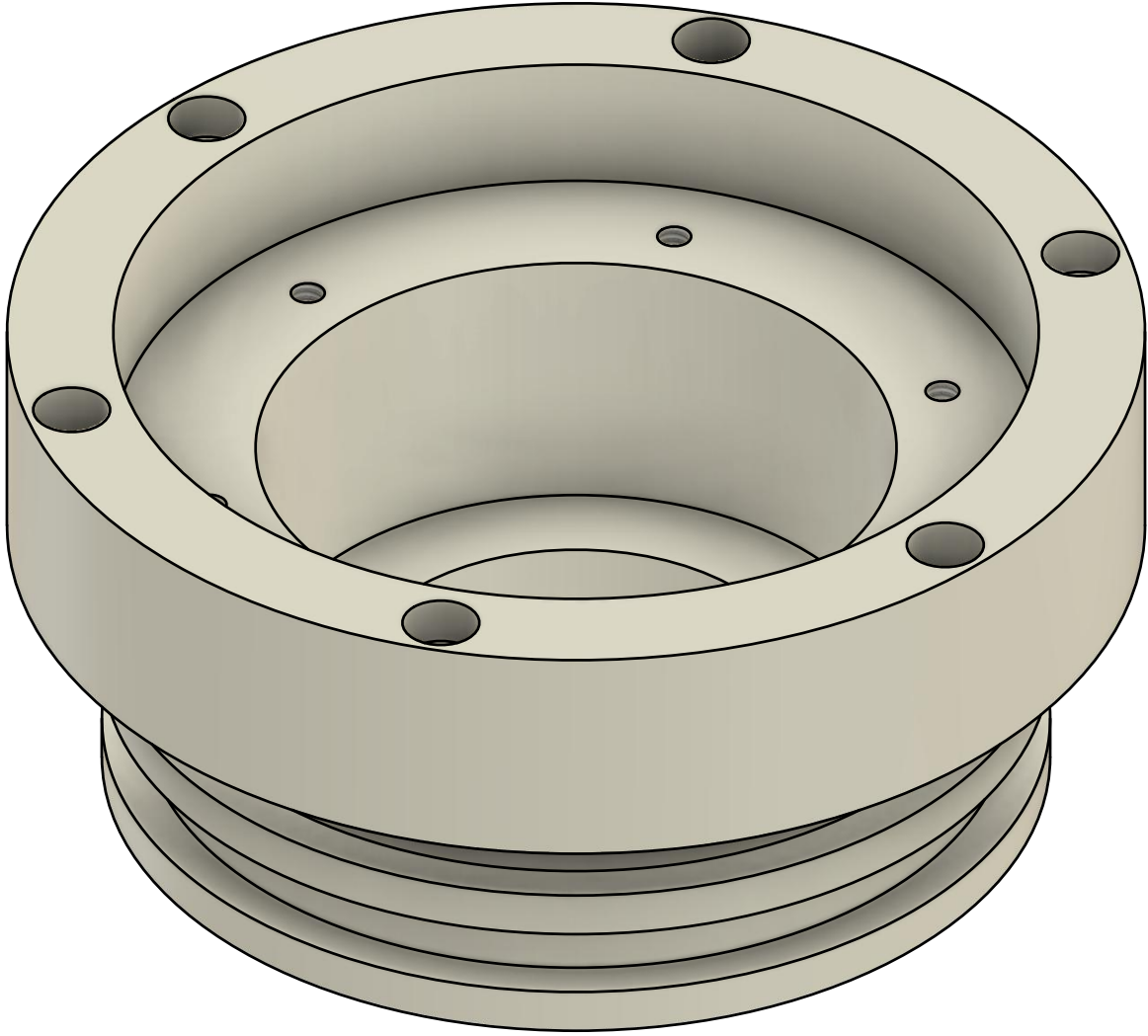
David Ruble

2024-06-13

SCALE 1:1

WEIGHT

SHEET 1/1



M: Acetal, black.

PROJECT

BHR

TITLE

Cap_Top

Rev 5: Top & Bot Caps same outside dimensions

APPROVED

SIZE

CODE

DWG NO

REV

CHECKED

A

5

DRAWN

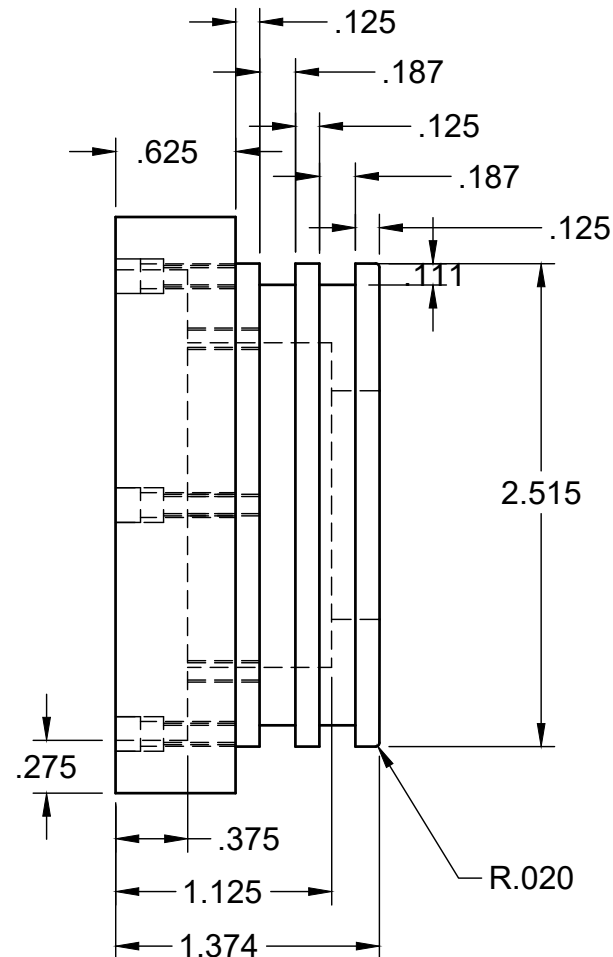
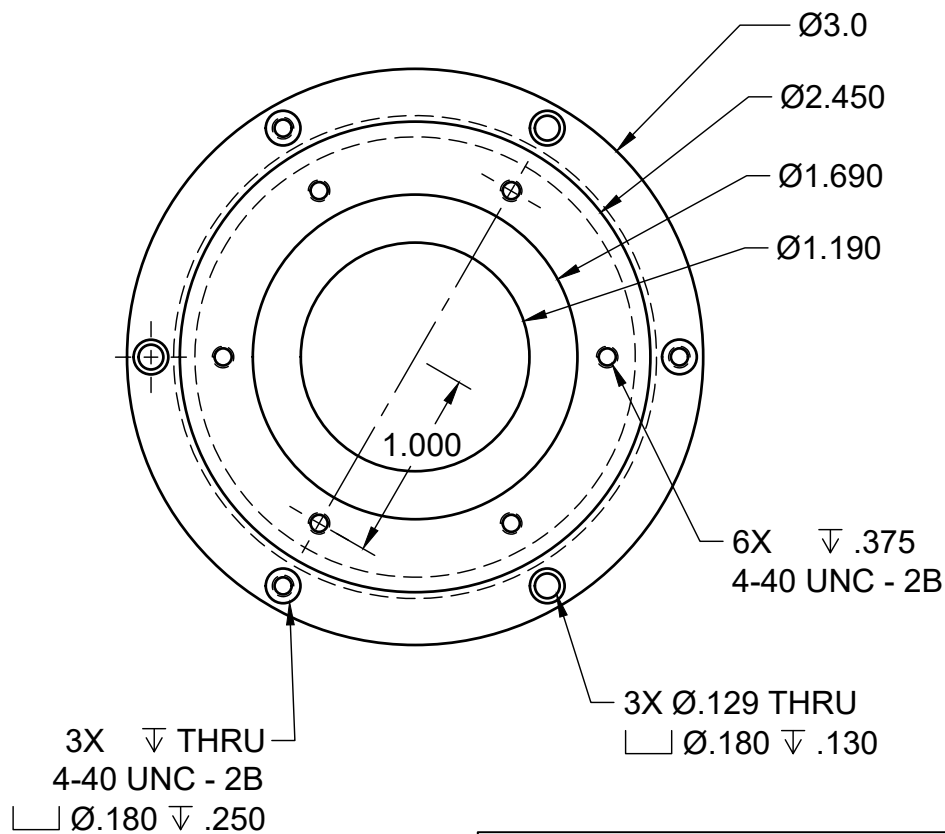
David Ruble

2024-04-17

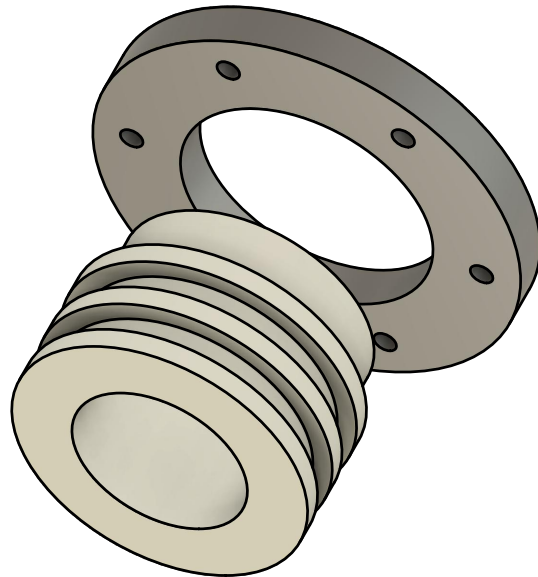
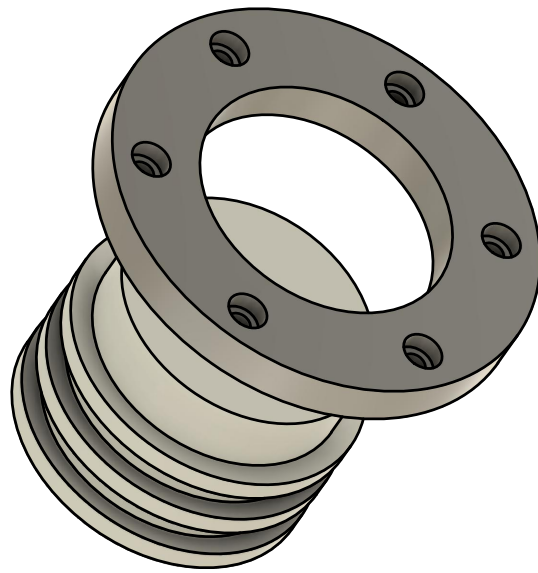
SCALE 1:0.5

WEIGHT

SHEET 1/1



M: acetal, black grooves for: o-ring -228 inner edge rounded, eased leading edge.		PROJECT			
		BHR			
		TITLE			
		Cap_Top			
		Rev 5+: Top & Bot Caps same outside dimensions			
APPROVED		SIZE	CODE	DWG NO	REV
CHECKED		A			6
DRAWN	David Ruble	2024-06-13	SCALE 1:1	WEIGHT	SHEET 1/1



irr White Delrin Diffuser
w/ irr retaining ring.

9mm thickness

PROJECT

BHR

TITLE

Window_irr_Diffuser_OV

APPROVED

CHECKED

SIZE

A

CODE

DWG NO

REV

DRAWN

David Ruble

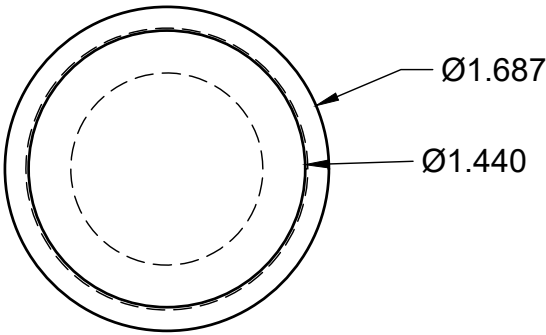
2024-01-18

SCALE 1:1

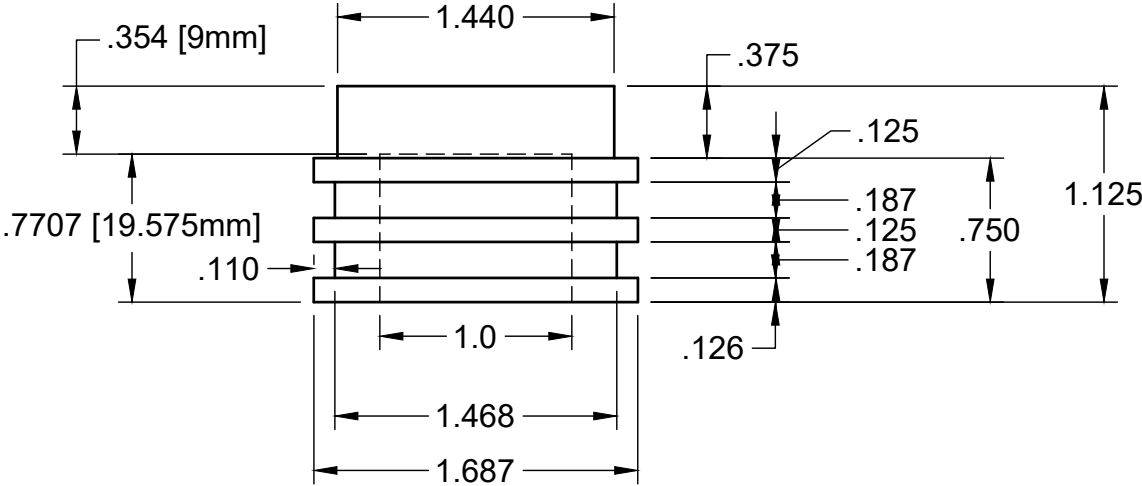
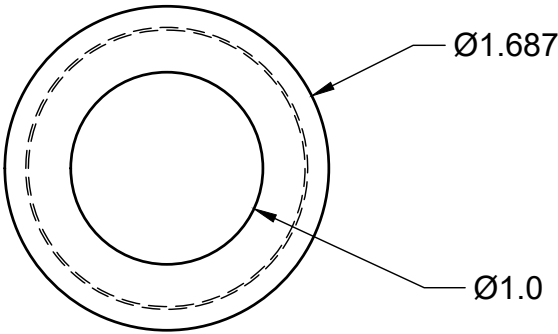
WEIGHT

SHEET 1/1

Top View: "the outside"



Inside View



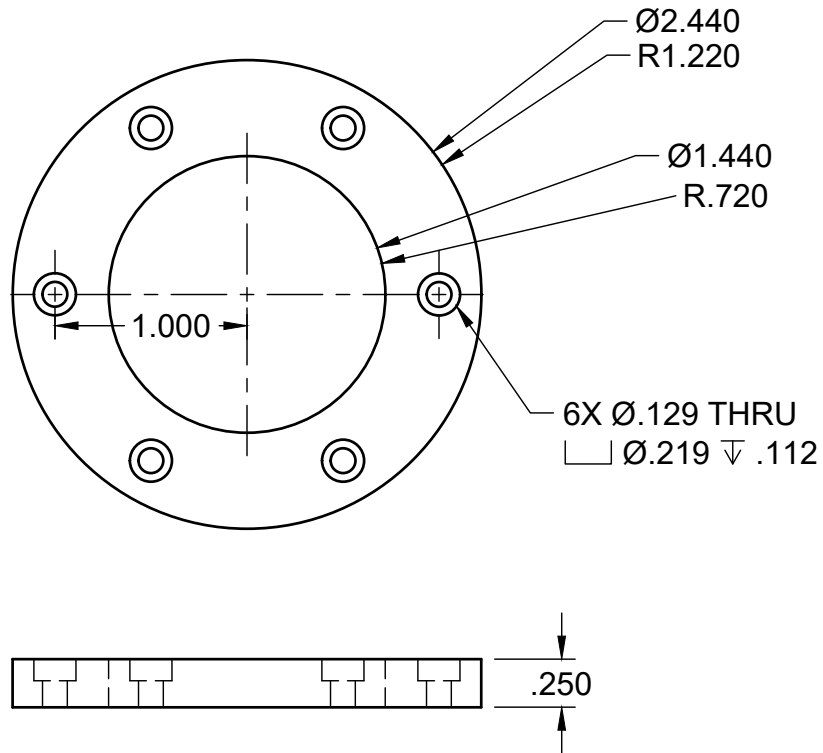
M: Delrin White
Diffuser Window

thickness: 9mm = ~0.3543"

bore depth = ~0.7707"

bore depth = ~19.575 mm

PROJECT			
BHR_irr			
TITLE			
White Window			
3 Side views			
APPROVED	SIZE	CODE	DWG NO
CHECKED	A		
DRAWN	David Ruble	2024-06-13	REV
SCALE	1:1	WEIGHT	3
SHEET		1/1	



M: milled from the $\frac{1}{4}$ " acetal sheet.

M: acetal, black
6x: Counter Bore for 4-40 SOC

wwi: white window irr
diffuser window

fits w/the irradiance white window

PROJECT

BHR

TITLE

wwi_Retaining_Ring

white window well irr; for the irradiance version

APPROVED

CHECKED

SIZE

A

CODE

DWG NO

REV

2

DRAWN

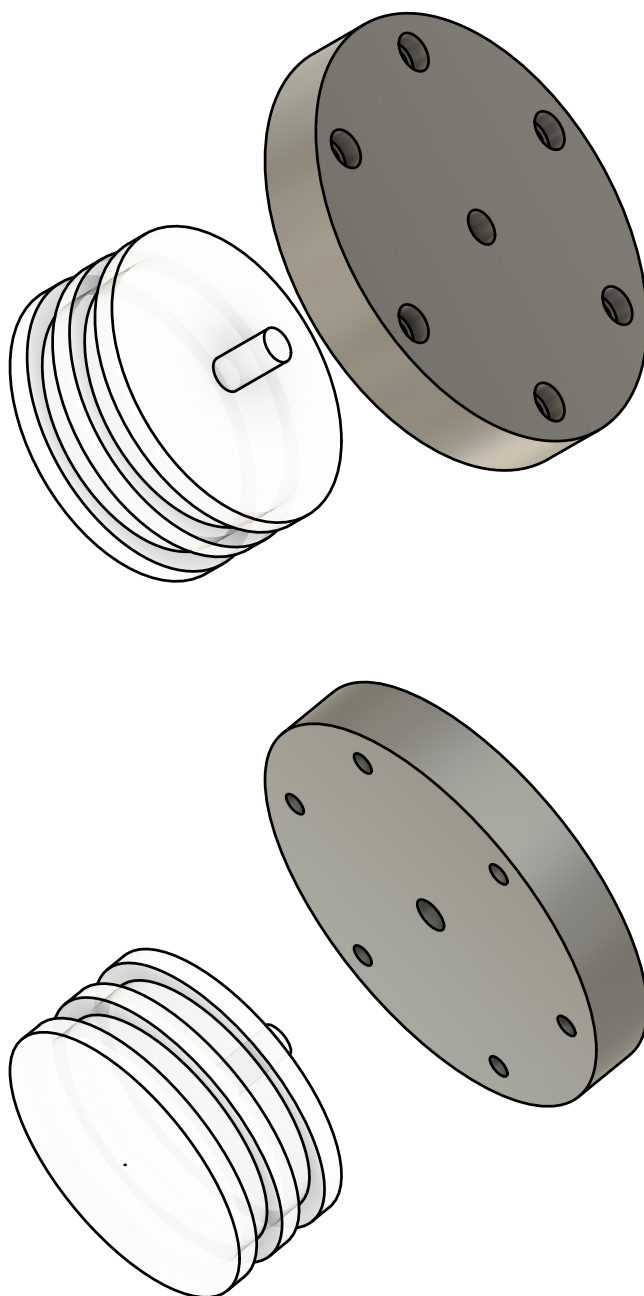
David Ruble

2023-06-13

SCALE 1:1

WEIGHT

SHEET 1/1



Acrylic O-Ring window
w/ Rad retaining ring.

5mm / FOV

PROJECT

BHR

TITLE

Window_Rad_Acrylic_OV

APPROVED

CHECKED

SIZE

A

CODE

DWG NO

REV

DRAWN

David Ruble

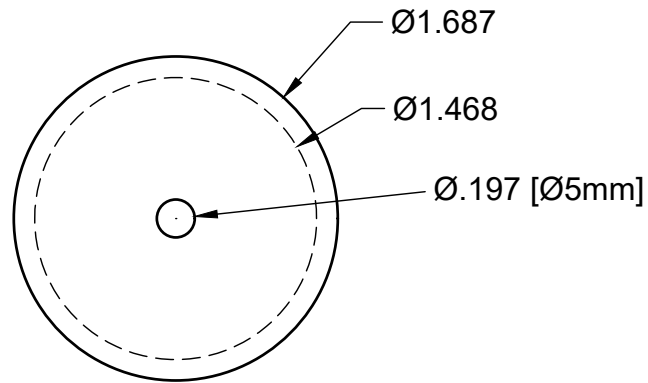
2023-07-13

SCALE 1:1

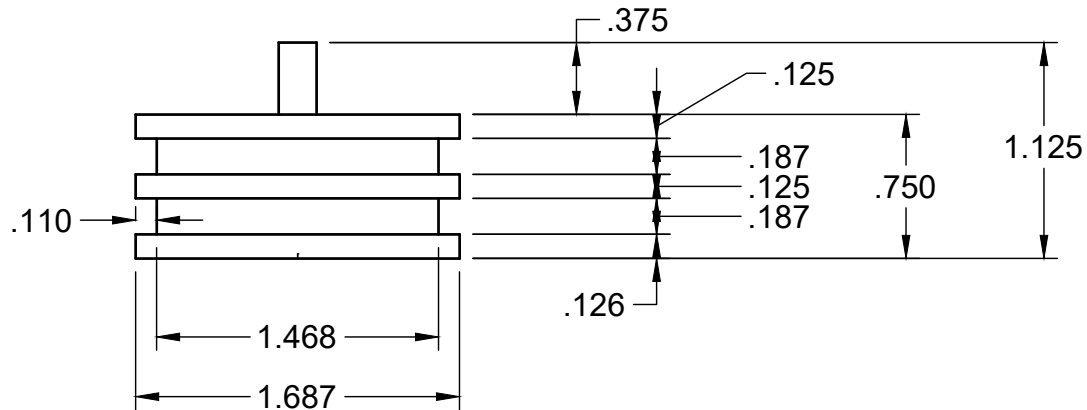
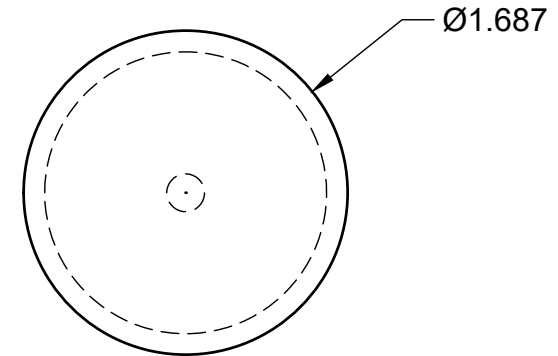
WEIGHT

SHEET 1/1

Top View:



Inside /bottom View:



M: clear Acrylic
Radiance Window
o-ring grooves for -221

PROJECT

BHR

TITLE

Window Acrylic Rad

Radiance Window: 3 Side view

APPROVED

CHECKED

DRAWN David Ruble 2043-06-13

SIZE

A

CODE

DWG NO

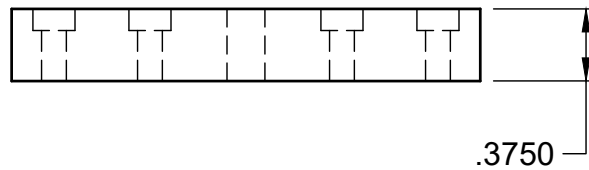
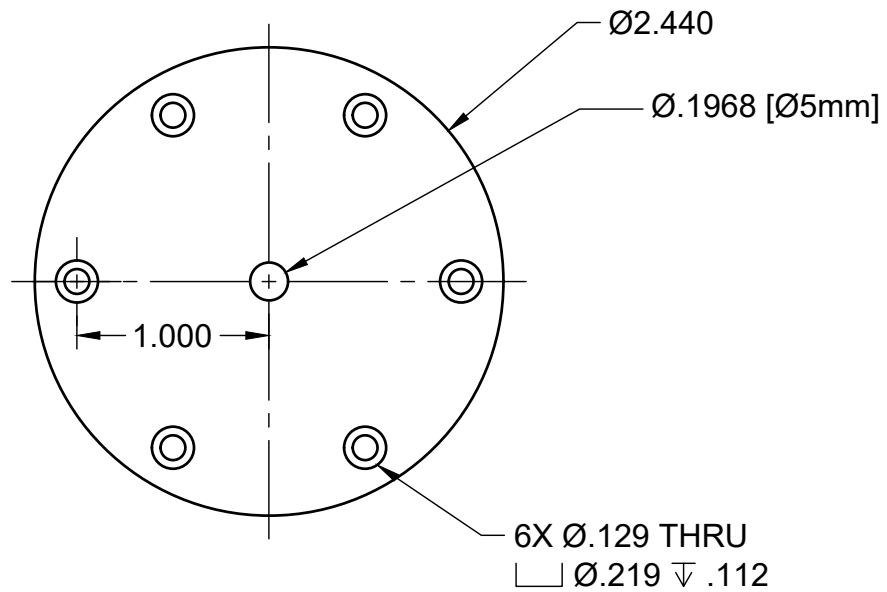
REV

2

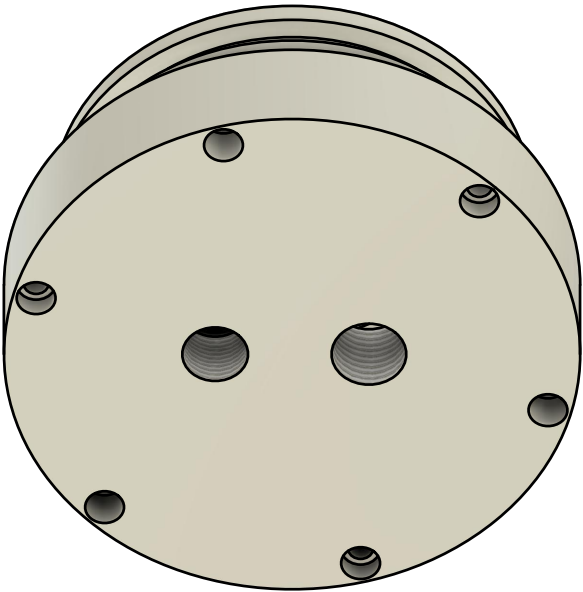
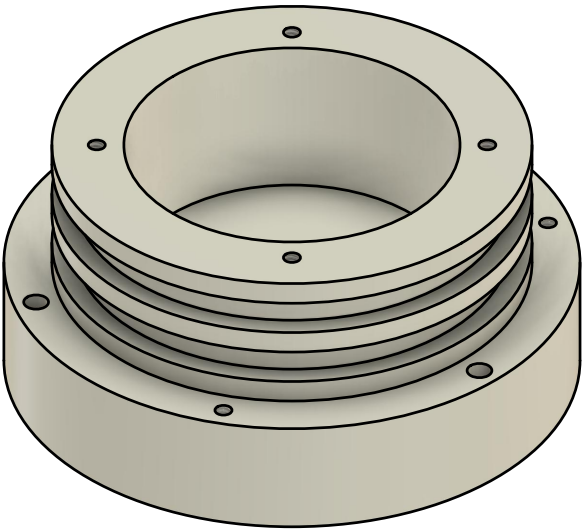
SCALE 1:1

WEIGHT

SHEET 1/1



M: acetal, black. 6x: Counter Bore for 4-40 SOC 5mm FOV Hole 5mm = 0.19685" fits w/the Radiance Acrylic window .			PROJECT BHR			
			TITLE wwa_Retaining_Ring_Rad			
acrylic window well			for the Radiance version			
APPROVED	SIZE	CODE	DWG NO			REV
CHECKED	A					3
DRAWN	David Ruble	2024-06-13	SCALE 1:1		WEIGHT	SHEET 1/1



M: Acetal, black.

3" Bottom Cap

w/ SubConn $\frac{7}{16}$ -20

w/ T-Plug $\frac{1}{8}$ -27 NPT

PROJECT

BHR

TITLE

Cap Bot "Bottom Cap"

OverView . Rev 5: duplicate Top Cap outside dimensions

APPROVED

CHECKED

SIZE

A

CODE

DWG NO

REV

5

DRAWN

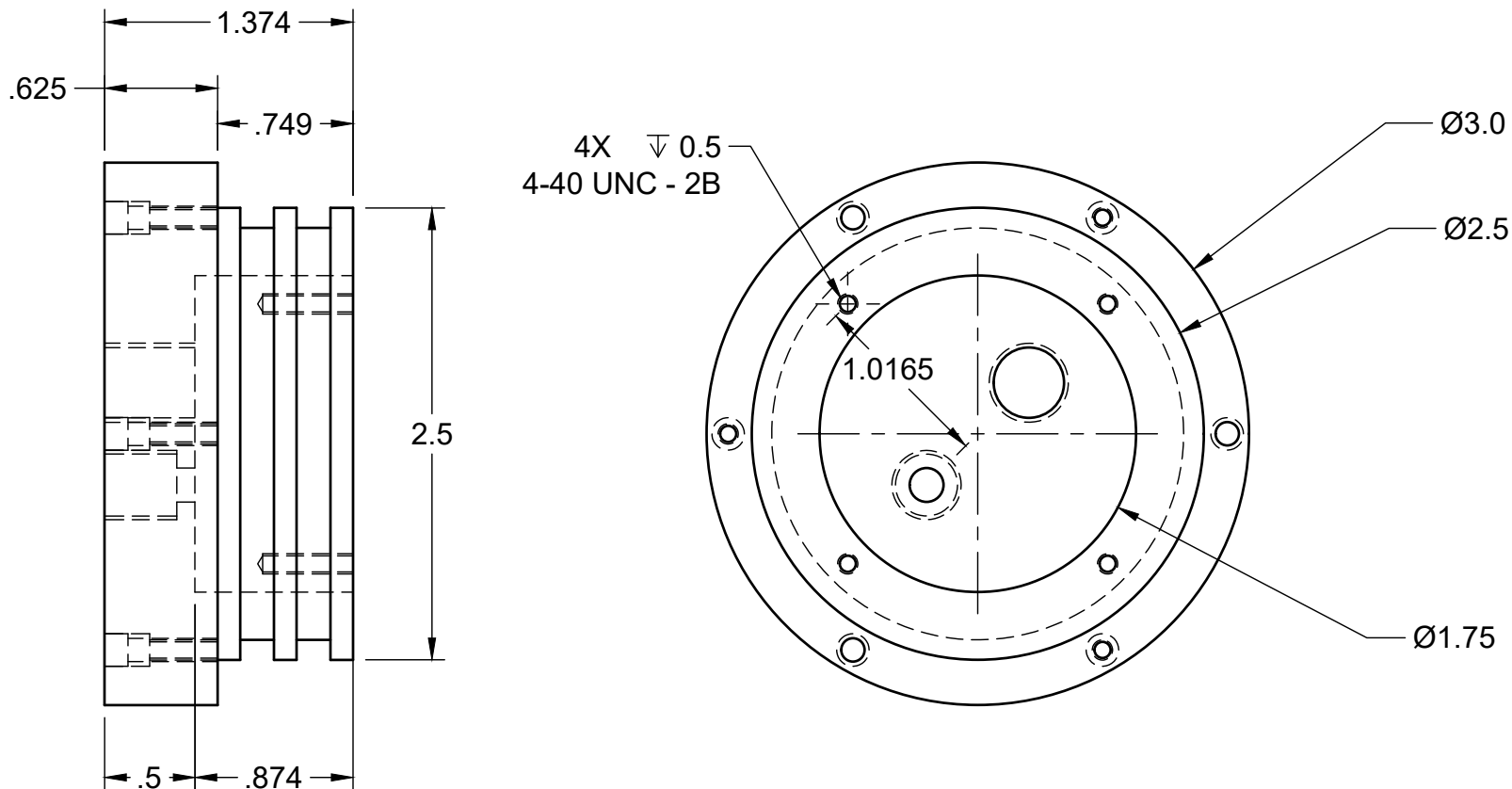
David Ruble

2024-04-17

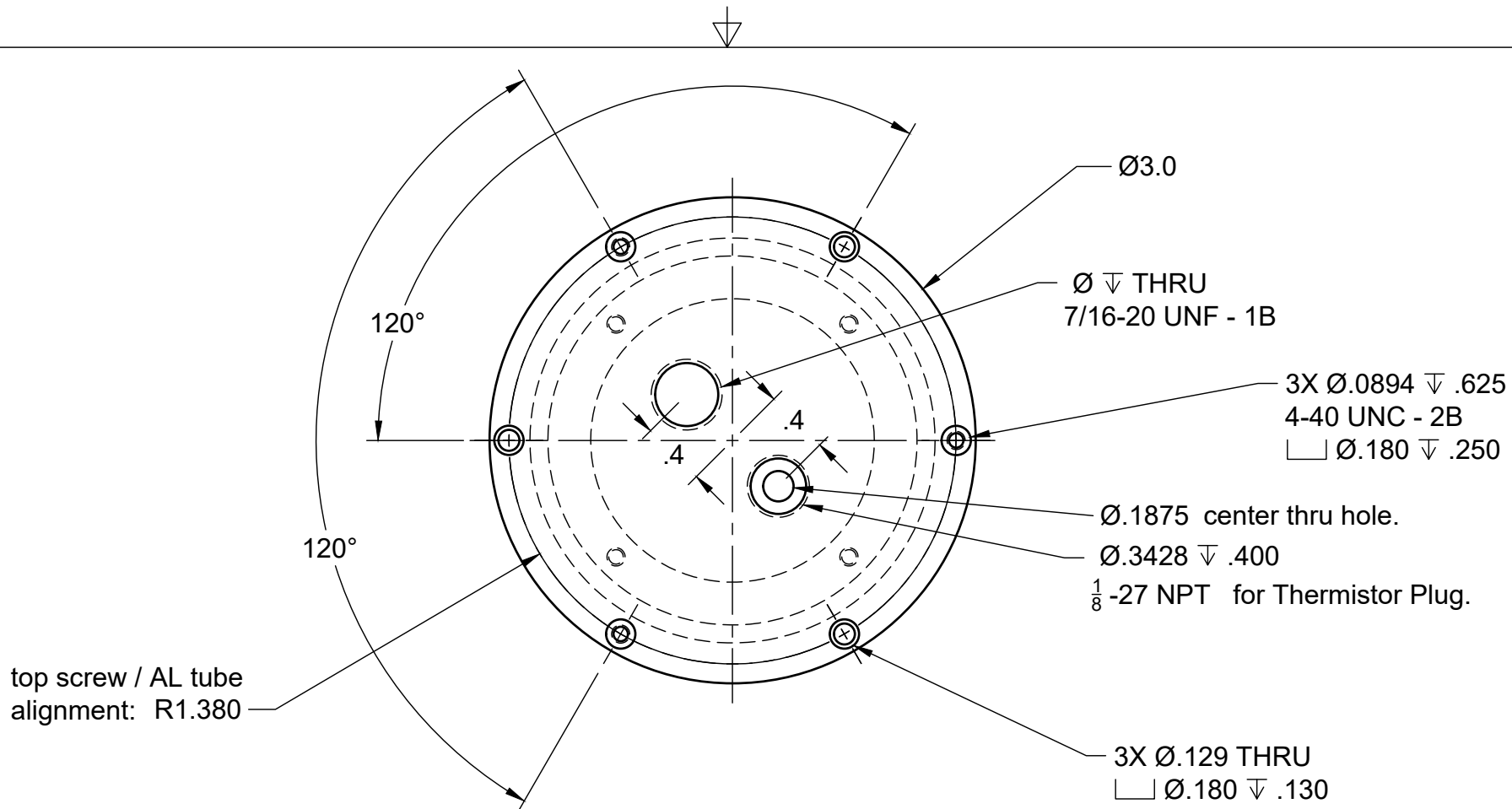
SCALE 1:1

WEIGHT

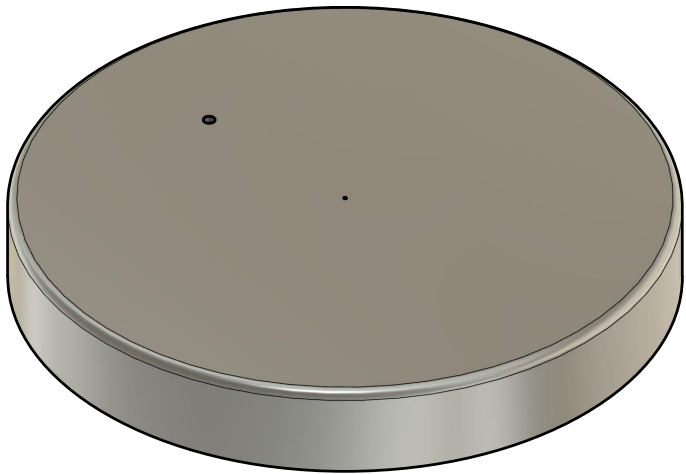
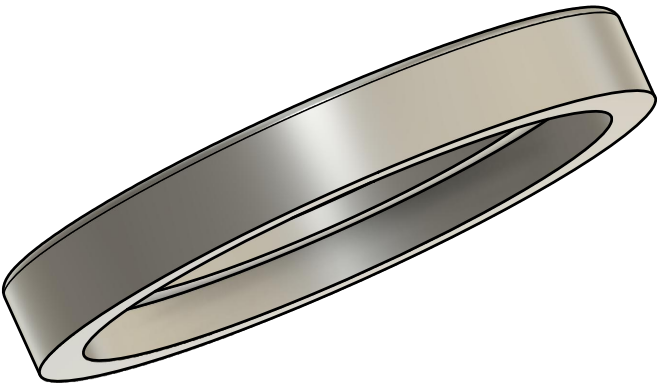
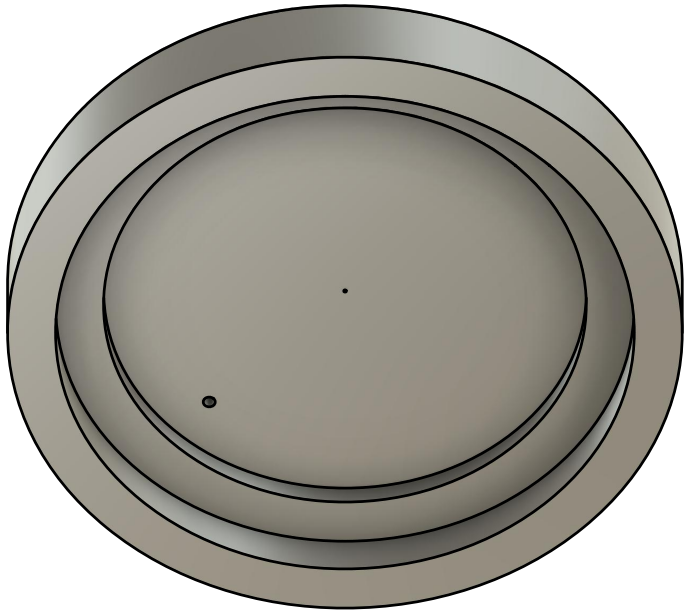
SHEET 1/1



the Bottom Bulkhead Cap/Lid the "In-Side" & side view.		PROJECT			
		BHR			
the 4x .5" deep 4-40 @ R 1.0165" for mount /match to the "inF_bot" ring. [internal Frame bottom ring.]		TITLE			
		Cap_Bot			
		InSide . Rev 5: duplicate Top Cap outside dimensions			
APPROVED		SIZE	CODE	DWG NO	REV
CHECKED		A			5
DRAWN	David Ruble	2024-06-13	SCALE 1:1	WEIGHT	SHEET 1/1



the Bottom Bulkhead Cap/Lid the "OutSide" view. Counter Bore (3x), #4 hex cap. & JackScrews (3x): 4-40 subConn: $\frac{7}{16}$ -20 T-Plug: $\frac{1}{8}$ -27 NPT		PROJECT			
		BHR			
		TITLE			
		Cap_Bot			
		OutSide. Rev 5: duplicate Top Cap outside dimensions			
APPROVED		SIZE	CODE	DWG NO	REV
CHECKED		A			5
DRAWN	David Ruble	2024-06-13	SCALE 1:1	WEIGHT	SHEET 1/1

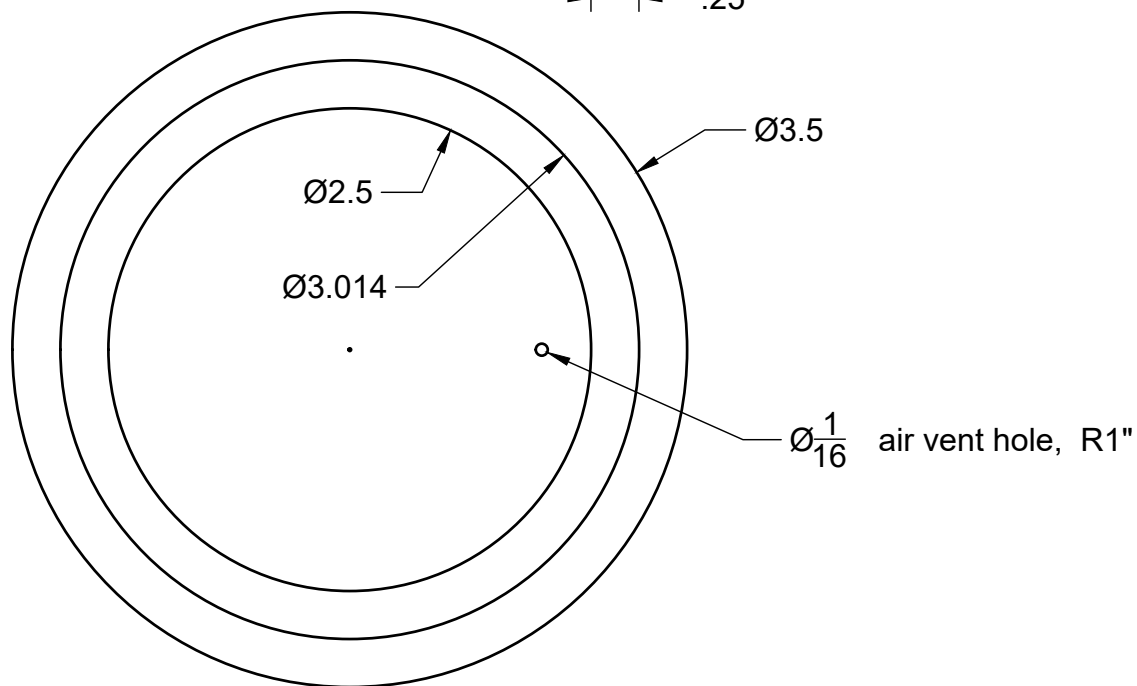
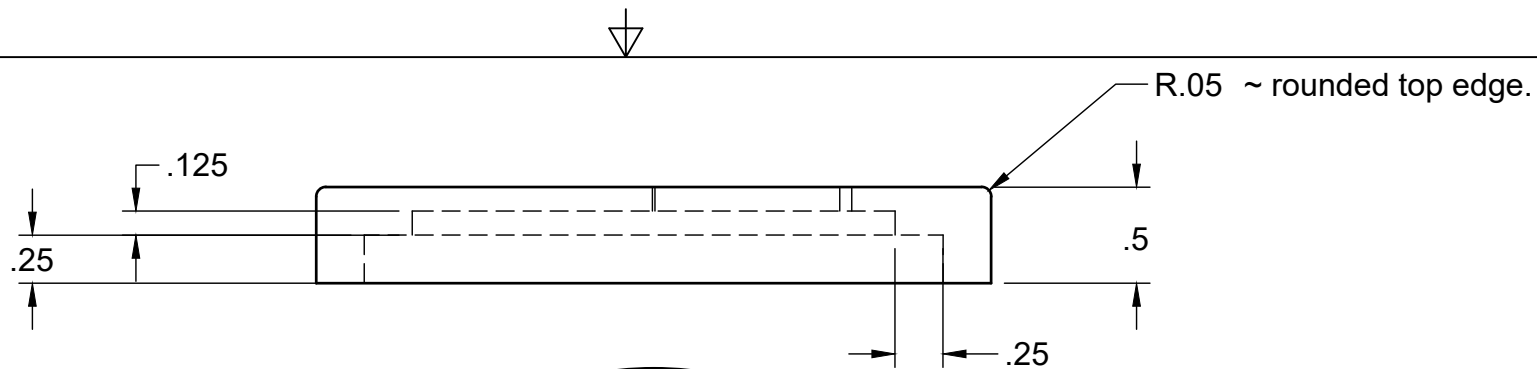


Cap Cover for 3" OD BHR

PROJECT
Cap_Covers

TITLE
Cover_Cap BHR 3" can

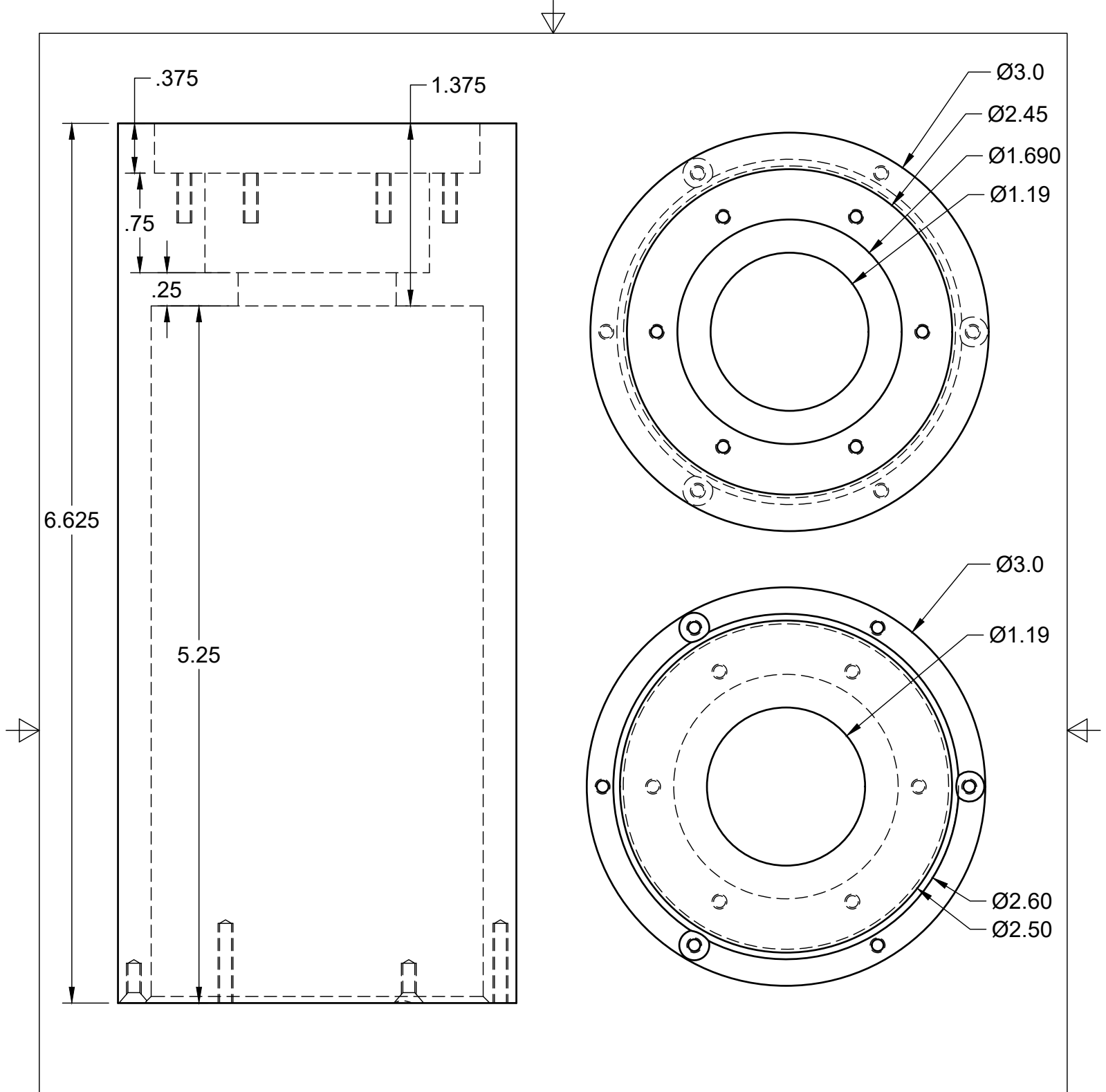
APPROVED			SIZE A	CODE	DWG NO	REV
CHECKED						
DRAWN	David Ruble	2024-01-18	SCALE 1:1		WEIGHT	SHEET 1/1



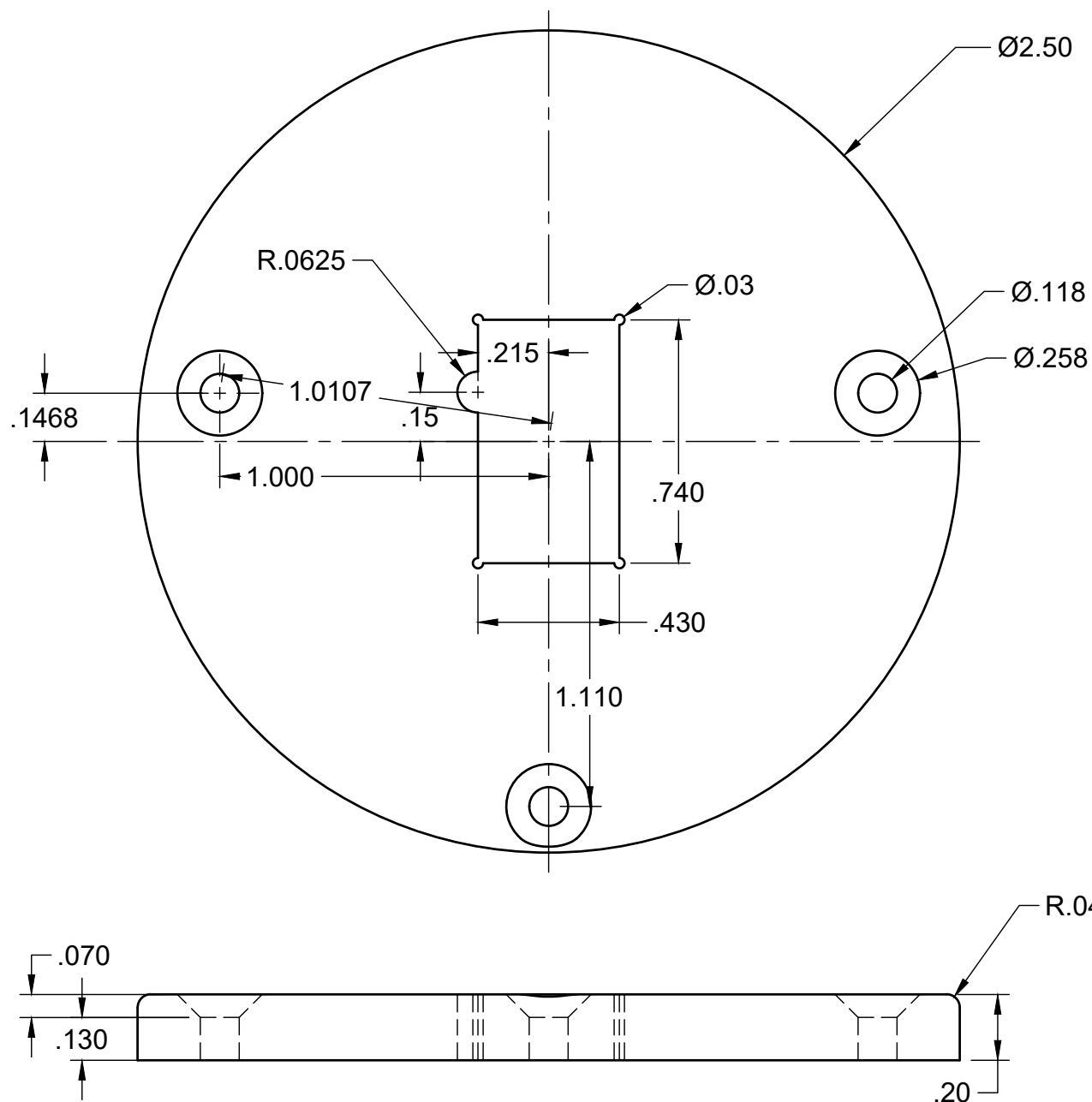
Cap Cover for Window		PROJECT Cap Covers			
		TITLE Cover_Cap BHR 3" can			
APPROVED	A	CODE	DWG NO	REV	2
CHECKED					
DRAWN	David Ruble	2022-07-20	SCALE 1:1	WEIGHT	SHEET 1/1



	PROJECT BHR				
	TITLE Ace Tube OD3 wTop Cap OV				
APPROVED	SIZE A	CODE	DWG NO		REV
CHECKED					2
DRAWN	David Ruble	2024-12-16	SCALE 1:1	WEIGHT	SHEET 1/1

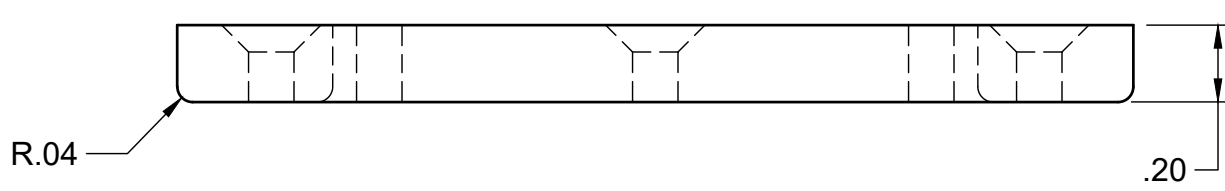
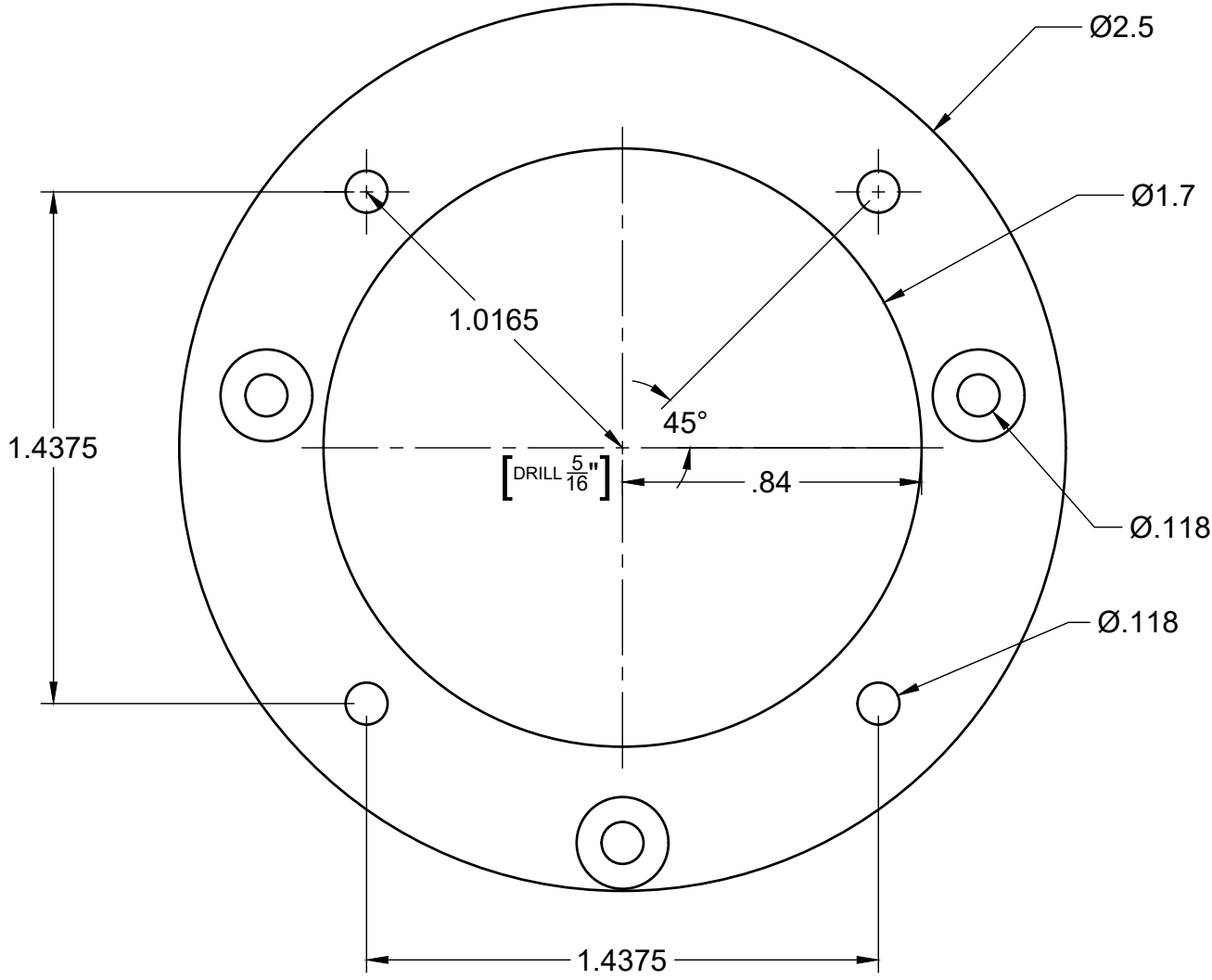


		PROJECT			
		BHR			
		TITLE			
		Ace Tube OD3 wTop Cap			
APPROVED		SIZE	CODE	DWG NO	REV
CHECKED		A			2
DRAWN	David Ruble	2024-12-16	SCALE 1:1	WEIGHT	SHEET 1/1

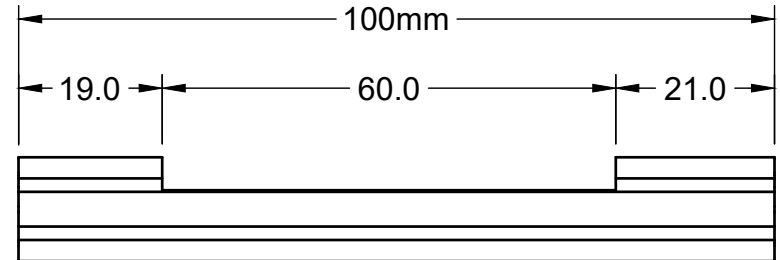
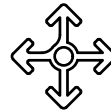
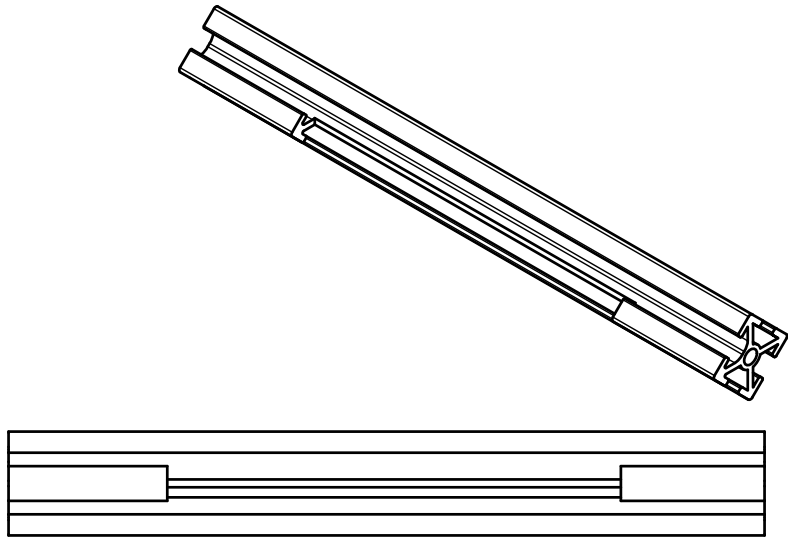


the "R.0625" small side knotch can be aligned w/the Y= 0.1468 holes, [=0.15 easier to sketch.]
and can be same Ø as the M3 screw, or a $\frac{1}{8}$ " is ok.

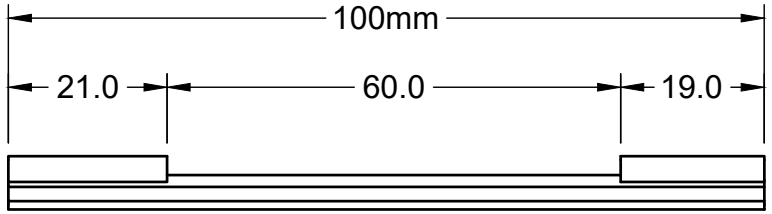
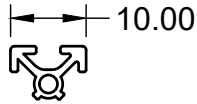
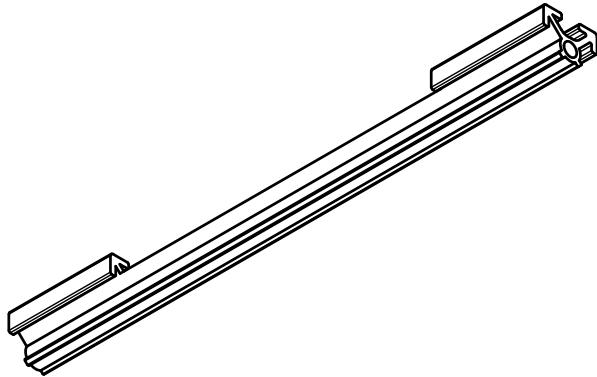
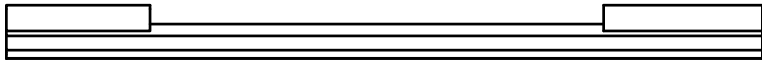
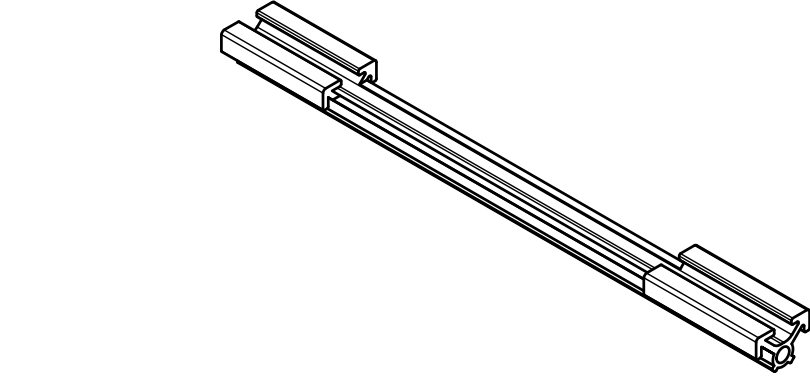
M: Black, Acetal		PROJECT		
3x Counter Sink for M3 screws		BHR		
2x holes, upper pair, R = 1.0107" and are 8.5° from horizontal.		TITLE		
R = 0.0625" side knotch.		inF_top Internal Frame Top		
		w/ dog bone; Rev 4: Thermistor cove; "small side knotch"		
APPROVED	SIZE	CODE	DWG NO	REV
CHECKED	A			4
DRAWN	David Ruble	2025-09-08	SCALE 2:1	WEIGHT
			SHEET 1/1	



Mounts on the Cap Bot. 3x Counter Sink for 3M screws & the 4x simple holes aligned with the "Cap Bot inside" holes.			PROJECT BHR			
			TITLE inF_bot			
			internal Frame Bottom mount to Bottom Cap			
APPROVED			SIZE	CODE	DWG NO	REV
CHECKED			A			3
DRAWN	David Ruble	2025-09-08	SCALE 1:0.5		WEIGHT	SHEET 1/1



	PROJECT				
	BHR				
	TITLE				
T-Slot cut : Upper rail					
Internal Frame Upper rails (2x)					
APPROVED	SIZE	CODE	DWG NO		REV
CHECKED	A				1
DRAWN	David Ruble	2024-12-13	SCALE 1:1	WEIGHT	SHEET 1/1



				PROJECT			
				BHR			
				TITLE			
				T Slot cut : bottom rail			
				Internal Frame bottom rail			
APPROVED		SIZE		CODE		DWG NO	
CHECKED		A					
DRAWN		David Ruble		2024-12-12		SCALE 1:1	
						WEIGHT	
						SHEET 1/1	